

Configuración del control de versiones con Git y GitHub

En esta sección se configura el sistema de control de versiones del proyecto utilizando **GitHub** como repositorio remoto para almacenar el código fuente del Sistema de Gestión Hospitalaria.

Instalación de Git

Actualizar los repositorios del sistema.

```
sudo apt update
```

Instalar Git.

```
sudo apt install git -y
```

Verificar la instalación.

```
git --version
```

Resultado esperado:

```
git version 2.49
```

Configuración inicial de Git

Configurar el nombre del usuario.

```
git config --global user.name "pcruz2891"
```

Configurar el correo asociado a GitHub.

```
git config --global user.email "pcruzr2891@gmail.com"
```

Verificar la configuración.

```
git config --list
```

Resultado esperado.

```
user.name=pcruz2891
```

```
user.email=pcruzr2891@gmail.com
```

Configuración de la autenticación mediante SSH

Generar la clave SSH.

```
ssh-keygen -t ed25519 -C "pcruzr2891@gmail.com"
```

Cuando aparezca:

Enter file in which to save the key (/home/pcruzr2891/.ssh/id_ed25519):

Presiona **Enter**.

Cuando solicite una contraseña para la clave:

Enter passphrase (empty for no passphrase):

Puedes dejarla vacía o establecer una contraseña para la clave SSH (no es la contraseña de GitHub).

Al finalizar aparecerá un mensaje similar a:

Your identification has been saved in /home/pcruzr2891/.ssh/id_ed25519

Your public key has been saved in /home/pcruzr2891/.ssh/id_ed25519.pub

```
Instalando Git...
git ya está en su versión más reciente (1:2.53.0-lubuntu1).
Los paquetes indicados a continuación se instalaron de forma automática y ya no son necesarios:
linux-headers-7.0.0-14 linux-image-unsigned-7.0.0-14-generic linux-modules-7.0.0-14-generic linux-tools-7.0.0-14-generic
linux-headers-7.0.0-14-generic linux-main-modules-zfs-7.0.0-14-generic linux-tools-7.0.0-14
Útilice «sudo apt autoremove» para eliminarlos.

Resumen:
  Actualizando: 0, Instalando 0, Eliminando 0, no actualizando: 19

Instalando Tree...
tree ya está en su versión más reciente (2.3.1-1).
Los paquetes indicados a continuación se instalaron de forma automática y ya no son necesarios:
linux-headers-7.0.0-14 linux-image-unsigned-7.0.0-14-generic linux-modules-7.0.0-14-generic linux-tools-7.0.0-14-generic
linux-headers-7.0.0-14-generic linux-main-modules-zfs-7.0.0-14-generic linux-tools-7.0.0-14
Útilice «sudo apt autoremove» para eliminarlos.

Resumen:
  Actualizando: 0, Instalando 0, Eliminando 0, no actualizando: 19

Versión instalada:
git version 2.53.0

Configuración de Git
Nombre de usuario: pcruzr2891
Correo de GitHub: pcruzr2891@gmail.com

Configuración realizada.

Generando clave SSH...
Generating public/private ed25519 key pair.
Enter file in which to save the key (/root/.ssh/id_ed25519):
Enter passphrase for "/root/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_ed25519
Your public key has been saved in /root/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:q1+Lu+KS6PGGK3jvw70BgGL+40c4FD7Yw7irAvwd4 pcruzr2891@gmail.com
The key's randomart image is:
+--[ED25519 256]--+
|
|..*..S
|+o*o..
|BBoO+...
|=B&E+.o.
|+B+X*X*o.
+----[SHA256]-----+
```

Iniciar el agente SSH

```
eval "$(ssh-agent -s)"
```

Agregar la clave.

```
ssh-add ~/.ssh/id_ed25519
```

Mostrar la clave pública

```
cat ~/.ssh/id_ed25519.pub
```

Obtendrás un resultado similar a:

```
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAA..... pcruzr2891@gmail.com
```

se debe Copias **todo** el contenido.

Registrar la clave en GitHub

Ingresa a:

<https://github.com>

Selecciona:

Settings

↓

SSH and GPG keys

↓

New SSH key

Completa:

Title

Ubuntu 24.04

Key

Pega la clave copiada anteriormente.

Haz clic en:

Add SSH key

**pcruz2891 (pcruz2891)**
Your personal account

[Go to your personal profile](#)

Public profile

Account

Appearance

Accessibility

Notifications

Access

Billing and licensing

Emails

Password and authentication

Sessions

SSH and GPG keys

Credentials

Organizations

Enterprises

Moderation

Code, planning, and automation

Repositories

Codespaces

Packages

SSH keys

New SSH key

This is a list of SSH keys associated with your account. Remove any keys that you do not recognize.

Authentication keys

 **sgb**
SHA256:FE1j1y3F1b0Rxc25EvCp6ouGxyHvD6gVec7gxRH21jell
Added on Jul 22, 2026
Last used within the last week — Read/write

Delete

Check out our guide to [connecting to GitHub using SSH keys](#) or troubleshoot [common SSH problems](#).

GPG keys

New GPG key

There are no GPG keys associated with your account.

Learn how to [generate a GPG key and add it to your account](#).

Vigilant mode

☐ **Flag unsigned commits as unverified**
This will include any commit attributed to your account but not signed with your GPG or S/MIME key.
Note that this will include your existing unsigned commits.
[Learn about vigilant mode](#)

Verificar la conexión

Ejecuta:

```
ssh -T git@github.com
```

La primera vez aparecerá:

Are you sure you want to continue connecting (yes/no/[fingerprint])?

Escribe:

```
Des:1 https://packages.microsoft.com/repos/code/stable/main amd64 code amd64 1.130.0-1784734578 [242 MB]
Des:2 http://cn.archive.ubuntu.com/ubuntu/resolute-updates/main amd64 socat amd64 1.8.1.1-1ubuntu0.1 [403 kB]
Descargados 243 MB en 17s (14.2 MB/s)
Preconfigurando paquetes ...
Seleccionando el paquete code previamente no seleccionado.
(Leyendo la base de datos ... 198534 ficheros o directorios instalados actualmente.)
Preparando para desempaquetar .../code_1.130.0-1784734578_amd64.deb ...
Desempaquetando code (1.130.0-1784734578) ...
Seleccionando el paquete socat previamente no seleccionado.
Preparando para desempaquetar .../socat_1.8.1.1-1ubuntu0.1_amd64.deb ...
Desempaquetando socat (1.8.1.1-1ubuntu0.1) ...
Configurando code (1.130.0-1784734578) ...
Configurando socat (1.8.1.1-1ubuntu0.1) ...
Procesando disparadores para desktop-file-utils (0.28-1build1) ...
Procesando disparadores para gnome-menus (3.38.1-1ubuntu1) ...
Procesando disparadores para man-db (2.13.1-1build1) ...
Procesando disparadores para shared-mime-info (2.4-1build3) ...

=====
versión instalada
=====
You are trying to start Visual Studio Code as a super user which isn't recommended. If this was intended, please add the argument '--no-sandbox' and specify an alternate user data directory using the '--user-data-dir' argument.
=====
Instalación finalizada correctamente
=====
root@sgb-M7: /#
```

yes

Resultado esperado.

Hi pcruz2891! You've successfully authenticated, but GitHub does not provide shell access.

```
=====
CLAVE SSH
=====
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAICVvuISMxYAFBqx6b4sNy73RtiM183PNGH8dnHU2I4Nd pcruzr2891@gmail.com

Copia esta clave.
Después agrégala en:
GitHub -> Settings -> SSH and GPG Keys

Presiona ENTER cuando hayas agregado la clave...

Verificando conexión...
The authenticity of host 'github.com (140.82.114.3)' can't be established.
ED25519 key fingerprint is: SHA256:+DiY3wvV6TujJhbpZisF/zLDA0zPMSvHdkr4UvCoqu
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
git@github.com: Permission denied (publickey).

=====
Nombre del proyecto: SistemaGestionHospitalaria
mkdir: SistemaGestionHospitalaria: File exists
Reinicializado el repositorio Git existente en /SistemaGestionHospitalaria/.git/
En la rama master
nada para hacer commit, el árbol de trabajo está limpio

=====
Proyecto creado correctamente.
=====

.
├── config
│   ├── kafka
│   ├── kong
│   ├── mongodb
│   ├── postgresql
│   ├── redis
│   └── zookeeper
├── docker
├── microservices
│   ├── ms-consulta
│   ├── ms-ehr
│   ├── ms-farmacia
│   └── ms-turnos
└── README.md

14 directories, 1 file
root@sgh-MV:/#
```

Create your first project

Ready to start building? Create a repository for a new idea or bring over an existing repository to keep contributing to it.

Create repository

Import repository

[Home](#)

Ask anything or type @ to add context

Ask

☰ All repositories

Auto

△

Agent

 Create issue

Write code

 Git

 Pull requests

Playlist

GitHub for beginners on YouTube

Designed to help you master the basics of GitHub, whether you're new to coding or looking to enhance your version control skills.

 [Start playlist](#)



Getting started

0/3 complete ...

- ▼ Create your first code project

It's your own space to experiment, learn, and share code with the world.

Create project

> [Start a Copilot chat](#)

> Complete your profile

Crear el repositorio en GitHub

Ingresa a GitHub.

Selecciona:

New Repository

The screenshot shows the GitHub 'New Repository' page for a repository named 'SistemaGestionHospitalaria', which is currently set to 'Public'. At the top, there are buttons for 'Pin', 'Watch' (0), 'Fork' (0), and 'Star' (0). Below these, there are two main sections: 'Start coding with Codespaces' and 'Add collaborators to this repository'. The 'Start coding with Codespaces' section includes a button 'Create a codespace'. The 'Add collaborators to this repository' section includes a button 'Invite collaborators'. Below these sections is a 'Quick setup' section with a dropdown menu for 'Set up in Desktop' or 'HTTPS' or 'SSH', and a text input field for the repository URL: 'https://github.com/pcruz2891/SistemaGestionHospitalaria.git'. Below the 'Quick setup' section is a section titled '...or create a new repository on the command line' with a code block containing the following commands:

```
echo "# SistemaGestionHospitalaria" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/pcruz2891/SistemaGestionHospitalaria.git
git push -u origin main
```

 Below this is another section titled '...or push an existing repository from the command line' with a code block containing the following commands:

```
git remote add origin https://github.com/pcruz2891/SistemaGestionHospitalaria.git
git branch -M main
git push -u origin main
```

 At the bottom of the page, there is a 'ProTip!' section with the text: 'Use the URL for this page when adding GitHub as a remote.'

Repository name

SistemaGestionHospitalaria

Description

Sistema de Gestión Hospitalaria basado en Microservicios.

Selecciona:

- Public (o Private)
- No agregar README
- No agregar .gitignore
- No agregar License

Haz clic en **Create repository**.

Crear el proyecto en Ubuntu

Ir al directorio de trabajo.

```
cd ~/Documentos
```

Crear el proyecto.

```
mkdir SistemaGestionHospitalaria
```

Entrar al proyecto.

```
cd SistemaGestionHospitalaria
```

Crear la estructura de carpetas

```
mkdir -p \  
microservices/ms-consulta \  
microservices/ms-ehr \  
microservices/ms-turnos \  
microservices/ms-farmacia \  
config/kong \  
config/kafka \  
config/postgresql \  
config/mongodb \  
config/redis \  
config/zookeeper \  
docker
```


Verificar la estructura.

tree

```
root@sgh-MV:/SistemaGestionHospitalaria# tree
.
├── config
│   ├── kafka
│   ├── kong
│   ├── mongodb
│   ├── postgresql
│   ├── redis
│   └── zookeeper
├── docker
├── microservices
│   ├── ms-consulta
│   ├── ms-ehr
│   ├── ms-farmacia
│   └── ms-turnos
└── README.md

14 directories, 1 file
root@sgh-MV:/SistemaGestionHospitalaria#
```

Inicializar Git

git init

```
14 directories, 1 file
root@sgh-MV:/SistemaGestionHospitalaria# git init
Reinicializado el repositorio Git existente en /SistemaGestionHospitalaria/.git/
root@sgh-MV:/SistemaGestionHospitalaria#
```

Crear el archivo README

touch README.md

Editar.

nano README.md

```
root@sgh-MV:/SistemaGestionHospitalaria# touch README.md
root@sgh-MV:/SistemaGestionHospitalaria#
```

Contenido.

Sistema de Gestión Hospitalaria

Proyecto desarrollado mediante una arquitectura de microservicios.

Guardar con:

Ctrl + O

Enter

Ctrl + X

Crear el archivo .gitignore

touch .gitignore

Editar.

nano .gitignore

Contenido.

Java

target/

Go

bin/

Visual Studio Code

.vscode/

```
bas:~ https://packages.microsoft.com/repos/code stable/main amd64 code amd64 1.130.0-1784734578 [242 MB]
bas:~ http://us.archive.ubuntu.com/ubuntu/resolute-updates/main amd64 socat amd64 1.8.1.1-1ubuntu0.1 [403 kB]
Descargados 243 MB en 17s (14.2 MB/s)
Preconfigurando paquetes ...
Seleccionando el paquete code previamente no seleccionado.
(Leyendo la base de datos ... 198534 ficheros o directorios instalados actualmente.)
Preparando para desempaquetar .../code_1.130.0-1784734578_amd64.deb ...
Desempaquetando code (1.130.0-1784734578) ...
Seleccionando el paquete socat previamente no seleccionado.
Preparando para desempaquetar .../socat_1.8.1.1-1ubuntu0.1_amd64.deb ...
Desempaquetando socat (1.8.1.1-1ubuntu0.1) ...
Configurando code (1.130.0-1784734578) ...
Configurando socat (1.8.1.1-1ubuntu0.1) ...
Procesando disparadores para desktop-file-utils (0.28-1build1) ...
Procesando disparadores para gnome-menus (3.38.1-1ubuntu1) ...
Procesando disparadores para man-db (2.13.1-1build1) ...
Procesando disparadores para shared-mime-info (2.4-5build3) ...

=====
Versión instalada
=====
You are trying to start Visual Studio Code as a super user which isn't recommended. If this was intended, please add the argument '--no-sandbox' and specify an alternate user data
directory using the '--user-data-dir' argument.
=====
Instalación finalizada correctamente
=====
root@ssh-MV: /#
```

Variables de entorno

.env

Logs

*.log

Guardar.

Verificar el estado del repositorio

git status

```
root@sgh-MV:/SistemaGestionHospitalaria# git status
En la rama develop
Tu rama está actualizada con 'origin/develop'.

Nada para hacer commit, el árbol de trabajo está limpio
root@sgh-MV:/SistemaGestionHospitalaria#
```

Agregar los archivos al repositorio

git add .

Crear el primer commit

git commit -m "Initial project structure"

Verificar el historial.

git log --oneline

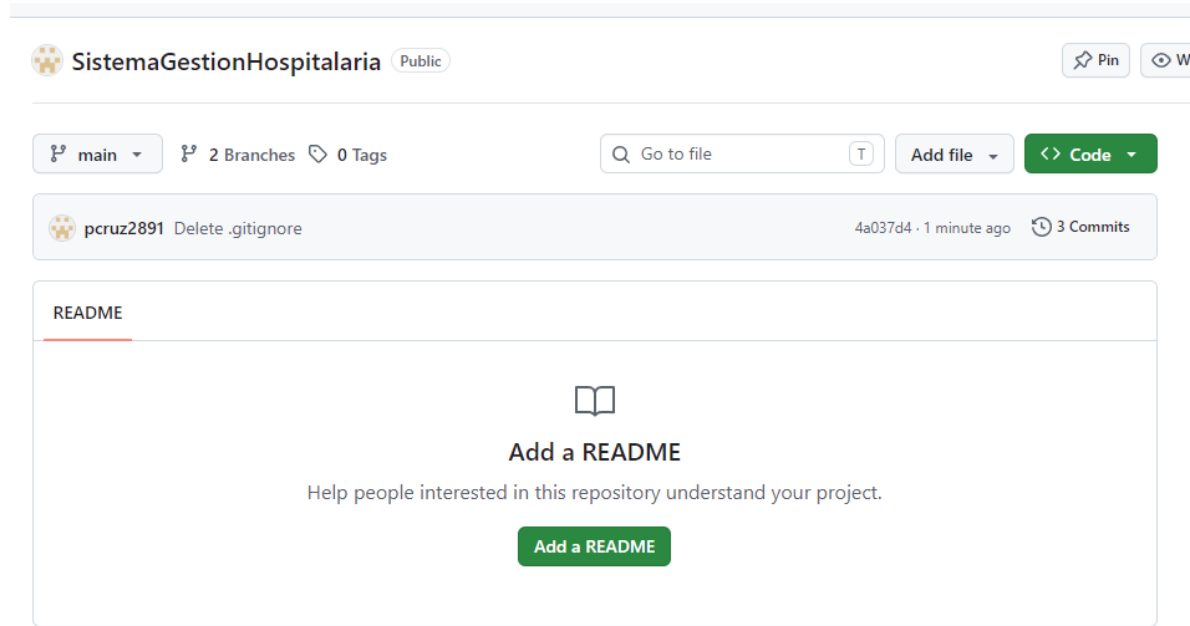
```
root@sgh-MV:/SistemaGestionHospitalaria# git log --oneline
ac6abd8 (HEAD -> develop, origin/main, origin/develop, main) Initial project structure
root@sgh-MV:/SistemaGestionHospitalaria#
```

Vincular el repositorio local con GitHub

Agregar el repositorio remoto utilizando la URL SSH.

git remote add origin [git@github.com:pcruz2891/SistemaGestionHospitalaria.git](https://github.com/pcruz2891/SistemaGestionHospitalaria.git)

Antes:



Despues

Verificar la configuración.

git remote -v

Resultado esperado.

origin git@github.com:pcruz2891/SistemaGestionHospitalaria.git (fetch)

origin git@github.com:pcruz2891/SistemaGestionHospitalaria.git (push)

```
root@sgh-MV:/SistemaGestionHospitalaria# git remote -v
origin git@github.com:pcruz2891/SistemaGestionHospitalaria.git (fetch)
origin git@github.com:pcruz2891/SistemaGestionHospitalaria.git (push)
root@sgh-MV:/SistemaGestionHospitalaria#
```

Publicar el proyecto

Renombrar la rama principal.

```
git branch -M main
```

Enviar el proyecto.

```
git push -u origin main
```

```
root@sgh-MV:/SistemaGestionHospitalaria# git push --force origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:pcruz2891/SistemaGestionHospitalaria.git
+ 4a037d4...ac6abd8 main -> main (forced update)
root@sgh-MV:/SistemaGestionHospitalaria#
```

Crear la rama develop

```
git checkout -b develop
```

Publicarla.

```
git push -u origin develop
```

The screenshot shows the GitHub profile page for user pcruz2891. On the left is a sidebar with navigation links: Public profile, Account, Appearance, Accessibility, Notifications, Access, Billing and licensing, Emails, Password and authentication, Sessions, SSH and GPG keys (highlighted), Credentials, Organizations, and Enterprises. The main content area is titled 'SSH keys' and includes a 'New SSH key' button. Below this is a list of authentication keys, showing one key named 'sgh' with its SHA256 fingerprint and a 'Delete' button. A link to a guide on connecting to GitHub using SSH keys is provided. Below the SSH keys section is the 'GPG keys' section, which has a 'New GPG key' button and states that there are no GPG keys associated with the account, with a link to learn how to generate and add a GPG key.

Crear ramas de funcionalidad

Microservicio Consulta.

```
git checkout develop
```

```
git checkout -b feature/ms-consulta
```

```
git push -u origin feature/ms-consulta
```

Microservicio EHR.

git checkout develop

git checkout -b feature/ms-ehr

git push -u origin feature/ms-ehr

Microservicio Turnos.

git checkout develop

git checkout -b feature/ms-turnos

git push -u origin feature/ms-turnos

Microservicio Farmacia.

git checkout develop

git checkout -b feature/ms-farmacia

git push -u origin feature/ms-farmacia

```
ssh-ed25519 AAAAC3NzaC1lZDIINTe5AAAAICVvuISMxyAFBqx6b4sNy13RtiM183PNGH8dnHU2I4Nd pcruzr2891@gmail.com
root@sgh-MV:/SistemaGestionHospitalaria# git commit -m "first commit"
En la rama main
nada para hacer commit, el árbol de trabajo está limpio
root@sgh-MV:/SistemaGestionHospitalaria# git branch -M main
root@sgh-MV:/SistemaGestionHospitalaria# git push -u origin main
Enumerando objetos: 4, listo.
Contando objetos: 100% (4/4), listo.
Compresión delta usando hasta 4 hilos
Comprimiendo objetos: 100% (2/2), listo.
Escribiendo objetos: 100% (4/4), 325 byte | 325.00 KiB/s, listo.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:pcruzr2891/SistemaGestionHospitalaria.git
 * [new branch]      main -> main
rama 'main' configurada para rastrear 'origin/main'.
root@sgh-MV:/SistemaGestionHospitalaria#
```

Verificar las ramas

Ramas locales.

git branch

Ramas remotas.

git branch -r

```
Everything up-to-date
root@sgh-MV:/SistemaGestionHospitalaria# touch microservices/ms-consulta/.gitkeep
touch microservices/ms-ehr/.gitkeep
touch microservices/ms-turnos/.gitkeep
touch microservices/ms-farmacia/.gitkeep

touch config/kafka/.gitkeep
touch config/kong/.gitkeep
touch config/postgresql/.gitkeep
root@sgh-MV:/SistemaGestionHospitalaria# touch microservices/ms-ehr/.gitkeep
touch config/mongodb/.gitkeep
root@sgh-MV:/SistemaGestionHospitalaria# touch microservices/ms-turnos/.gitkeep
touch config/zookeeper/.gitkeep
root@sgh-MV:/SistemaGestionHospitalaria# touch microservices/ms-farmacia/.gitkeep

touch docker/.gitkeeproot@sgh-MV:/SistemaGestionHospitalaria#
root@sgh-MV:/SistemaGestionHospitalaria# touch config/kafka/.gitkeep
root@sgh-MV:/SistemaGestionHospitalaria# touch config/kong/.gitkeep
root@sgh-MV:/SistemaGestionHospitalaria# touch config/postgresql/.gitkeep
root@sgh-MV:/SistemaGestionHospitalaria# touch config/mongodb/.gitkeep
root@sgh-MV:/SistemaGestionHospitalaria# touch config/redis/.gitkeep
root@sgh-MV:/SistemaGestionHospitalaria# touch config/zookeeper/.gitkeep
root@sgh-MV:/SistemaGestionHospitalaria#
root@sgh-MV:/SistemaGestionHospitalaria# touch docker/.gitkeep
root@sgh-MV:/SistemaGestionHospitalaria# git add .
git commit -m "Agregar estructura inicial del proyecto"
root@sgh-MV:/SistemaGestionHospitalaria# git commit -m "Agregar estructura inicial del proyecto"
git push origin main[main 14073bf] Agregar estructura inicial del proyecto
11 files changed, 0 insertions(+), 0 deletions(-)
create mode 100644 config/kafka/.gitkeep
create mode 100644 config/kong/.gitkeep
create mode 100644 config/mongodb/.gitkeep
create mode 100644 config/postgresql/.gitkeep
create mode 100644 config/redis/.gitkeep
create mode 100644 config/zookeeper/.gitkeep
create mode 100644 docker/.gitkeep
create mode 100644 microservices/ms-consulta/.gitkeep
create mode 100644 microservices/ms-ehr/.gitkeep
create mode 100644 microservices/ms-farmacia/.gitkeep
create mode 100644 microservices/ms-turnos/.gitkeep
root@sgh-MV:/SistemaGestionHospitalaria# git push origin main
Enumerando objetos: 7, listo.
Contando objetos: 100% (7/7), listo.
Compresión delta usando hasta 4 hilos
Comprimiendo objetos: 100% (4/4), listo.
Escribiendo objetos: 100% (6/6), 602 byte | 602.00 KiB/s, listo.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:pcruz2891/SistemaGestionHospitalaria.git
   ac6abd8..14073bf  main -> main
root@sgh-MV:/SistemaGestionHospitalaria#
```

Validación de que se Subió todo a GitHub

<https://github.com/pcruz2891/SistemaGestionHospitalaria>

 SistemaGestionHospitalaria Public Pin 1

main 1 Branch 0 Tags Add file Code

 pcruz2891 · Agregar estructura inicial del proyecto 14073bf · 1 minute ago 2 Commits

config	Agregar estructura inicial del proyecto	1 minute ago
docker	Agregar estructura inicial del proyecto	1 minute ago
microservices	Agregar estructura inicial del proyecto	1 minute ago
.gitignore	Initial project structure	3 hours ago
README.md	Initial project structure	3 hours ago

README 

SistemaGestionHospitalaria

```
root@sgh-MV:/SistemaGestionHospitalaria# tree
.
├── config
│   ├── kafka
│   ├── kong
│   ├── mongodb
│   ├── postgresql
│   ├── redis
│   └── zookeeper
├── docker
├── microservices
│   ├── ms-consulta
│   ├── ms-ehr
│   ├── ms-farmacia
│   └── ms-turnos
└── README.md

14 directories, 1 file
root@sgh-MV:/SistemaGestionHospitalaria#
```